

1. What is the core intuition behind a Support Vector Machine classifier?

- A. Minimize the number of misclassified points only
- B. Separate the classes with the widest possible margin
- C. Fit a probability distribution to each class
- D. Maximize the number of support vectors

2. In SVM terminology, the gap fitted between the two classes is often described as the:

- A. bucket
- B. street
- C. tube
- D. kernel

3. Which training instances determine the SVM decision boundary?

- A. All training instances equally
- B. Only the support vectors
- C. Only the instances farthest from the boundary
- D. A random subset chosen during training

4. For which kind of datasets are SVMs particularly well suited?

- A. Very large datasets with millions of instances
- B. Complex small- and medium-sized datasets
- C. Only perfectly linearly separable data
- D. Datasets with no feature scaling needed

5. A support vector is a training instance that lies:

- A. Far from the decision boundary
- B. On the edge of the street (the margin) or violates it
- C. Exactly at the center of its class cluster
- D. Always misclassified by the model

6. Adding more training instances 'off the street' (beyond the margin) will:

- A. Always widen the margin
- B. Not affect the decision boundary at all
- C. Always shift the boundary toward them
- D. Force a retraining from scratch

7. Which task is NOT something a basic SVM is described as being able to perform?

- A. Linear classification
- B. Nonlinear classification
- C. Regression
- D. Generating new synthetic samples

8. SVMs are described as being suited to outlier detection in addition to:

- A. clustering and dimensionality reduction
- B. classification and regression

- C. reinforcement learning
- D. time-series forecasting only

9. A good SVM decision boundary stays:

- A. as close as possible to the densest class
- B. as far as possible from the closest points of each class
- C. exactly equidistant from the two class centroids
- D. aligned with the first principal component

10. In the iris example from the lecture, the linear SVM is used to detect:

- A. Iris setosa from sepal measurements
- B. Iris virginica from petal length and width
- C. Iris versicolor from petal area
- D. all three species simultaneously

11. Why are SVMs sensitive to feature scales?

- A. They use a probabilistic loss that requires normalization
- B. An axis with a much larger scale dominates the width of the street
- C. The kernel trick only works on scaled data
- D. Scaling is required for the bias term to exist

12. Which Scikit-Learn transformer is recommended to scale features before an SVM?

- A. MinMaxScaler only
- B. StandardScaler
- C. OneHotEncoder
- D. PolynomialFeatures

13. What happens to the 'widest street' when one feature has a far larger scale than another?

- A. It becomes wider and more accurate
- B. It is dominated by the large-scale axis and looks unreasonable
- C. It is unaffected because SVMs are scale-invariant
- D. It rotates to align with the small-scale axis

14. By default, do SVM classifiers (SVC/LinearSVC) output class probabilities?

- A. Yes, always
- B. No, not by default
- C. Only with the RBF kernel
- D. Only for binary problems

15. The decision function of a linear SVM computes:

- A. a softmax over class scores
- B. $w^T x + b$
- C. the Euclidean distance to each centroid
- D. the entropy of the feature vector

16. A linear SVM predicts the positive class when:

- A. $w^T x + b < 0$
- B. $w^T x + b \geq 0$
- C. $\|w\| > 1$
- D. $b > 0$

17. The decision boundary of a linear SVM is the set of points where:

- A. $w^T x + b = 1$
- B. $w^T x + b = 0$
- C. $w^T x + b = -1$
- D. $\|w\| = 0$

18. For a problem with n features, the linear SVM decision boundary is:

- A. an n -dimensional sphere
- B. an $(n-1)$ -dimensional hyperplane
- C. a single point
- D. always a straight line in 2D only

19. The dashed margin lines of a linear SVM correspond to where the decision function equals:

- A. 0 and 1
- B. +1 and -1
- C. +0.5 and -0.5
- D. the class probabilities

20. In the LinearSVC iris pipeline, what is placed before the classifier?

- A. A PolynomialFeatures step
- B. A StandardScaler step
- C. A PCA step
- D. Nothing; scaling is unnecessary

21. Hard-margin classification requires that:

- A. some instances may lie inside the street
- B. every instance is off the street and on the correct side
- C. exactly half the instances are support vectors
- D. the margin width is fixed in advance

22. A major limitation of hard-margin classification is that it:

- A. only works if the data is linearly separable
- B. cannot use the kernel trick
- C. requires probabilistic outputs
- D. ignores feature scaling

23. Hard-margin SVMs are described as being:

- A. robust to outliers
- B. very sensitive to outliers — one point can wreck the boundary

- C. immune to feature scaling
- D. faster than soft-margin SVMs

24. Soft-margin classification seeks to balance:

- A. training speed against memory use
- B. a wide street against limited margin violations
- C. precision against recall directly
- D. the number of features against instances

25. In soft-margin classification, instances are allowed to:

- A. never cross the margin
- B. lie inside the street or on the wrong side
- C. be removed from the training set automatically
- D. change their labels

26. Compared with hard margin, soft margin usually:

- A. overfits more on real data
- B. generalizes much better on real data
- C. produces a narrower street always
- D. ignores support vectors

27. The C hyperparameter in an SVM controls the trade-off between:

- A. learning rate and epochs
- B. margin width and margin violations
- C. kernel degree and gamma
- D. training and test error directly

28. A low value of C tends to produce:

- A. a narrower margin with fewer violations
- B. a wider margin tolerating more violations
- C. no margin at all
- D. a probabilistic output

29. A high value of C tends to produce:

- A. a wider margin with more violations
- B. a narrower margin with fewer violations, risking overfitting
- C. a smoother RBF boundary
- D. an automatically scaled dataset

30. If your SVM is overfitting the training data, a reasonable first step is to:

- A. increase C
- B. reduce C
- C. remove the scaler
- D. switch to a hard margin

31. Lower C is often associated with:

- A. worse generalization
- B. better generalization
- C. guaranteed zero training error
- D. infinite margin width

32. In the soft-margin formulation, the slack variable ζ_i measures:

- A. the kernel value at instance i
- B. how much instance i is allowed to violate the margin
- C. the learning rate for instance i
- D. the probability of class i

33. The soft-margin objective minimizes $\frac{1}{2}w^Tw$ plus a term proportional to:

- A. the sum of slack variables, scaled by C
- B. the number of features
- C. the inverse of the margin
- D. the kernel matrix trace

34. Both the hard-margin and soft-margin SVM training problems are examples of:

- A. non-convex optimization
- B. convex Quadratic Programs (QP)
- C. linear programs only
- D. unconstrained problems

35. Why is $\frac{1}{2}\|w\|^2$ minimized rather than $\|w\|$ directly?

- A. $\frac{1}{2}\|w\|^2$ is smaller
- B. $\frac{1}{2}\|w\|^2$ is differentiable everywhere (derivative w), while $\|w\|$ is not at 0
- C. $\|w\|$ cannot be computed
- D. they give different optima

36. The margin width of a linear SVM is related to $\|w\|$ such that:

- A. a larger $\|w\|$ widens the street
- B. a smaller $\|w\|$ widens the street
- C. $\|w\|$ has no effect on margin
- D. margin = $\|w\|^2$

37. In the training-objective notation, the target t_i is defined as:

- A. 0 for negatives and 1 for positives
- B. +1 for positive instances and -1 for negative instances
- C. the raw feature value
- D. the slack variable

38. The hard-margin constraint can be written as:

- A. $t_i(w^Tx_i + b) \geq 1$ for all i
- B. $t_i(w^Tx_i + b) \leq 0$ for all i
- C. $w^Tx_i + b = t_i$
- D. $\|w\| = 1$

39. One way to handle nonlinearly separable data with a linear model is to:

- A. remove the nonlinear features
- B. add polynomial features (e.g., PolynomialFeatures)
- C. always use a hard margin
- D. disable feature scaling

40. In the make_moons example, what is added before LinearSVC to capture nonlinearity?

- A. RBF similarity features
- B. PolynomialFeatures
- C. a second scaler
- D. a probability calibrator

41. A drawback of adding explicit polynomial features at high degree is that it:

- A. always underfits
- B. explodes the number of features, making the model slow
- C. removes the need for scaling
- D. prevents overfitting entirely

42. A drawback of using a very low polynomial degree is that it:

- A. overfits complex data
- B. cannot fit complex datasets well
- C. creates too many features
- D. requires the kernel trick

43. The polynomial kernel lets an SVM behave as if many polynomial features were added, while:

- A. actually computing all of them explicitly
- B. not actually computing them
- C. ignoring the data entirely
- D. only working in 2D

44. In SVC, which kernel argument selects the polynomial kernel?

- A. kernel='rbf'
- B. kernel='poly'
- C. kernel='linear'
- D. kernel='sigmoid'

45. In the polynomial kernel SVC, the 'degree' hyperparameter controls:

- A. the regularization strength
- B. the polynomial order
- C. the reach of each instance
- D. the number of support vectors

46. If a polynomial-kernel SVM is overfitting, you might:

- A. increase the degree

- B. reduce the degree
- C. increase coef0 only
- D. remove C

47. If a polynomial-kernel SVM is underfitting, you might:

- A. reduce the degree
- B. increase the degree
- C. remove the scaler
- D. set C to zero

48. The coef0 hyperparameter of the polynomial kernel controls:

- A. the learning rate
- B. how much the model is influenced by high-degree vs low-degree terms
- C. the number of instances
- D. the slack penalty

49. The general form of the polynomial kernel is:

- A. $\exp(-\gamma \|a-b\|^2)$
- B. $(\gamma a^T b + r)^d$
- C. $\tanh(\gamma a^T b + r)$
- D. $a^T b$

50. A recommended practical strategy for tuning kernel hyperparameters is to:

- A. use a single fine grid search
- B. use a coarse grid search first, then refine around the best values
- C. never tune them
- D. tune only on the test set

51. In the polynomial features example on moons, which estimator is used as the final classifier?

- A. SVC(kernel='rbf')
- B. LinearSVC(loss='hinge')
- C. SGDRegressor
- D. LogisticRegression

52. Adding polynomial features as a preprocessing step works:

- A. only with SVMs
- B. with any model, not just SVMs
- C. only with tree-based models
- D. only after the kernel trick

53. In the polynomial kernel comparison plot, increasing degree from 3 to 10 mainly:

- A. smooths the boundary
- B. makes the boundary fit more complex/wiggly shapes
- C. removes support vectors
- D. disables regularization

54. The polynomial kernel is preferable to explicit PolynomialFeatures mainly because it:

- A. always generalizes better
- B. avoids the combinatorial blow-up in the number of features
- C. needs no hyperparameters
- D. produces probabilities

55. Similarity features measure:

- A. the distance between two classes
- B. how much each instance resembles particular landmarks
- C. the variance of each feature
- D. the slack of each instance

56. The Gaussian RBF similarity function $\phi_{\gamma}(x, \ell)$ is:

- A. $\gamma \cdot x^T \ell$
- B. $\exp(-\gamma \cdot \|x - \ell\|^2)$
- C. $\tanh(\gamma \cdot x^T \ell)$
- D. $(x - \ell)^2$

57. The Gaussian RBF similarity equals its maximum value when:

- A. x is far from the landmark
- B. x equals the landmark ℓ
- C. γ is zero
- D. x is at the origin

58. As an instance moves far from a landmark, its Gaussian RBF similarity:

- A. grows without bound
- B. approaches 0
- C. stays constant at 1
- D. becomes negative

59. The simplest choice of landmarks for similarity features is to:

- A. place one at the class centroids
- B. place a landmark at every training instance
- C. use a single landmark at the origin
- D. place landmarks randomly off the data

60. A downside of placing a landmark at every instance is that it creates:

- A. too few features
- B. m features for m instances (large feature count)
- C. negative similarity values
- D. non-convex optimization

61. In SVC, which argument selects the Gaussian RBF kernel?

- A. `kernel='poly'`
- B. `kernel='rbf'`
- C. `kernel='linear'`

D. kernel='sigmoid'

62. The RBF kernel's γ (gamma) hyperparameter sets:

- A. the regularization penalty
- B. the reach of each instance / how fast similarity decays
- C. the polynomial degree
- D. the number of support vectors

63. A large value of γ produces:

- A. a wide bell and smoother boundary
- B. a narrow bell and a wiggly, irregular boundary (can overfit)
- C. no boundary
- D. a linear boundary

64. A small value of γ produces:

- A. a narrow bell and irregular boundary
- B. a wide bell and a smoother boundary (can underfit)
- C. an infinite margin
- D. probabilistic outputs

65. Both γ and C in an RBF SVM act as:

- A. kernel selectors
- B. regularizers — reduce them if overfitting, raise them if underfitting
- C. learning rates
- D. scaling factors

66. If an RBF SVM is overfitting, a sensible adjustment is to:

- A. increase γ and C
- B. reduce γ and/or C
- C. remove the scaler
- D. switch to hard margin

67. If an RBF SVM is underfitting, a sensible adjustment is to:

- A. reduce γ and C
- B. increase γ and/or C
- C. remove the kernel
- D. add slack variables

68. The RBF kernel achieves the effect of many similarity features while:

- A. computing them all explicitly
- B. not actually computing them (the kernel trick)
- C. ignoring the data
- D. only using two landmarks

69. An RBF is defined as any function that depends only on:

- A. the dot product of inputs

- B. the distance between the input and a fixed point
- C. the sign of the inputs
- D. the number of features

70. In the lecture's γ/C grid, moving down a column (increasing C) tends to produce:

- A. more margin violations
- B. fewer margin violations
- C. wider bells
- D. fewer support vectors only via γ

71. A kernel function $K(a,b)$ computes:

- A. the class label of a and b
- B. a high-dimensional dot product without performing the mapping ϕ
- C. the Euclidean distance only
- D. the slack of the pair

72. For the 2nd-degree polynomial mapping ϕ , the identity $\phi(a)^T \phi(b)$ equals:

- A. $a^T b$
- B. $(a^T b)^2$
- C. $\exp(-\|a-b\|^2)$
- D. $\|a-b\|$

73. The kernel trick replaces every dot product $x_i^T x_{\square}$ in the dual with:

- A. $\|x_i - x_{\square}\|$
- B. $K(x_i, x_{\square})$
- C. $t_i t_{\square}$
- D. $\alpha_i \alpha_{\square}$

74. Mercer's theorem guarantees that if K satisfies Mercer's conditions, then:

- A. K is always linear
- B. some mapping ϕ exists with $K(a,b)=\phi(a)^T \phi(b)$
- C. the SVM cannot overfit
- D. ϕ is always finite-dimensional

75. Mercer's conditions require the kernel to be:

- A. differentiable and bounded
- B. continuous and symmetric
- C. positive and linear
- D. sparse and convex

76. Which is the linear kernel?

- A. $K(a,b)=a^T b$
- B. $K(a,b)=(\gamma a^T b + r)^d$
- C. $K(a,b)=\exp(-\gamma \|a-b\|^2)$
- D. $K(a,b)=\tanh(\gamma a^T b + r)$

77. Which is the Gaussian RBF kernel?

- A. $K(a,b)=a^T b$
- B. $K(a,b)=(\gamma a^T b + r)^d$
- C. $K(a,b)=\exp(-\gamma \|a-b\|^2)$
- D. $K(a,b)=\tanh(\gamma a^T b + r)$

78. Which is the sigmoid kernel?

- A. $K(a,b)=a^T b$
- B. $K(a,b)=(\gamma a^T b + r)^d$
- C. $K(a,b)=\exp(-\gamma \|a-b\|^2)$
- D. $K(a,b)=\tanh(\gamma a^T b + r)$

79. Every SVM primal problem has an equivalent:

- A. closed-form normal equation
- B. dual form with the same solution (here)
- C. unconstrained gradient descent form
- D. probabilistic interpretation

80. The dual SVM problem depends on the data only through:

- A. the slack variables
- B. dot products $x_i^T x_j$
- C. the bias term b
- D. the feature scales

81. From the dual solution, the primal weights are recovered as:

- A. $w = \sum \alpha_i t_i x_i$
- B. $w = \sum x_i$
- C. $w = b \cdot \sum t_i$
- D. $w = K(x, x)$

82. In the recovered weight expression, α_i is nonzero only for:

- A. misclassified points
- B. support vectors
- C. the bias term
- D. every instance

83. The dual form is faster than the primal when:

- A. the number of instances is much larger than features
- B. the number of instances is smaller than the number of features
- C. there are no support vectors
- D. the data is linearly separable

84. The dot-product structure of the dual is important because it:

- A. removes the need for a bias
- B. is what makes the kernel trick possible
- C. guarantees a unique global maximum margin

D. eliminates slack variables

85. For a kernelized SVM, the weight vector w may live in:

- A. a 2-dimensional space always
- B. a huge or even infinite-dimensional space that can't be formed explicitly
- C. the same space as the labels
- D. a space with one dimension per class

86. The Gaussian RBF kernel corresponds to a feature space that is:

- A. 1-dimensional
- B. finite and small
- C. infinite-dimensional
- D. equal to the number of classes

87. For a kernelized SVM, predictions $\hat{h}(x)$ require computing kernels between x and:

- A. all training instances
- B. only the support vectors
- C. the class centroids
- D. the test set

88. The kernelized prediction has the form $\hat{h}(x) = \sum \alpha_i t_i K(x_i, x) + b$, summed over:

- A. all instances
- B. the support vectors only
- C. the features
- D. the margins

89. The cost of making a kernelized SVM prediction scales with:

- A. the total number of training instances
- B. the number of support vectors
- C. the number of features only
- D. the number of classes

90. The linear SVM primal cost $J(w,b)$ is $\frac{1}{2}w^T w + C \sum \max(0, 1 - t_i(w^T x_i + b))$. The $\max(0, \cdot)$ term is called the:

- A. log loss
- B. hinge loss
- C. squared loss
- D. cross-entropy

91. The hinge loss $\max(0, 1 - t)$ becomes exactly zero once:

- A. $t \leq 0$
- B. $t \geq 1$
- C. $t = 0$
- D. $t < -1$

92. The hinge loss is not differentiable at $t = 1$, but gradient descent can still use:

- A. the second derivative
- B. any subderivative in $[-1, 0]$
- C. a finite-difference approximation only
- D. the kernel matrix

93. For online or out-of-core linear SVM training, one can minimize J with:

- A. the normal equation
- B. SGDClassifier (slower to converge than QP)
- C. a hard-margin solver
- D. grid search

94. In SVM regression, the objective is essentially flipped to:

- A. separate the classes as far apart as possible
- B. fit as many instances as possible inside the street (an ϵ -tube)
- C. maximize the number of support vectors
- D. minimize the bias only

95. In SVM regression, the width of the tube is controlled by:

- A. C
- B. ϵ (epsilon)
- C. γ
- D. the degree

96. Points that fall inside the ϵ -tube in SVM regression:

- A. dominate the fit
- B. do not affect the fit (the model is ϵ -insensitive)
- C. become support vectors automatically
- D. are discarded from training

97. A smaller ϵ in SVM regression tends to produce:

- A. a wider tube and fewer support vectors
- B. a narrower tube and more support vectors
- C. no support vectors
- D. a linear-only fit

98. Which Scikit-Learn class performs linear SVM regression?

- A. LinearSVC
- B. LinearSVR
- C. SVC
- D. SGDClassifier

99. For nonlinear SVM regression with a polynomial kernel, you would use:

- A. LinearSVR
- B. SVR(kernel='poly', ...)
- C. SVC(kernel='poly', ...)
- D. LinearSVC

100. Regarding training complexity, LinearSVC (liblinear, no kernel trick) scales approximately as:

- A. $O(m^2 \times n)$
- B. $O(m \times n)$ — almost linearly
- C. $O(m^3 \times n)$
- D. $O(n^3)$